Road-supported effective constitutive coupling and anisotropic routing responses in urban mobility

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Abstract

Conservation constrains the source–sink balance of urban movement but does not uniquely determine its internal routing. Here we identify an effective linear coupling between these two aspects of collective mobility. Using Hodge decomposition, we separate observed flows into source-determined and divergence-free components and estimate their relation within a road-supported routing space. Across Porto, San Francisco and Chengdu, the inferred operator recovers day-specific routing variations on held-out dates, with distinct strengths across cities. The operator exhibits pronounced, spatially heterogeneous directional selectivity that is not attributable to road geometry alone. Road support captures routing organization beyond rank-matched random subspaces, and the measured road arrangement remains distinguishable from matched non-road media; mobility-inferred media add information not specified by roads. Together, these findings establish an interpretable framework for identifying a road-supported effective coupling between source–sink organization and directional transport that is partial and observation-conditioned, rather than a regime-invariant constitutive law.

Cities organize movement as well as activities. The location of homes, jobs and services determines where journeys begin and end, while the arrangement of streets shapes the spaces through which those journeys pass. Neighbourhoods with similar net inflows can nevertheless experience different through-movement, and a redistribution of activity need not produce a proportional redistribution of traffic. Gravity, radiation and spatial-interaction models describe flows between places [1, 2, 3, 4, 5]; traffic assignment links demand to routes through travel costs and behavioural rules [6, 7]; and graph-based learning offers flexible descriptions of traffic and assignment [8, 9]. Between aggregate interaction and individual routes lies a useful urban-scale object: the local direction and intensity of collective movement.

Field descriptions make this intermediate scale accessible. Recurrent commuting can be represented by a vector field whose potential identifies mobility basins [10]; trajectory-based formulations extend the approach beyond commuting [11]; and Laplacian evolution identifies urban source–sink patterns [12]. Hodge decomposition separates a potential-driven component, associated with the imbalance between arrivals and departures, from circulation that redistributes movement without changing that balance [13, 14]. The two sectors are geometrically orthogonal, but spatial orthogonality does not preclude reproducible statistical coupling. Related field studies identify scaling, temporal response, directional organization and anomalous movement [15, 16, 17, 18]. Whether changes in one sector systematically accompany changes in the other is the question that turns a descriptive decomposition into an empirical relation.

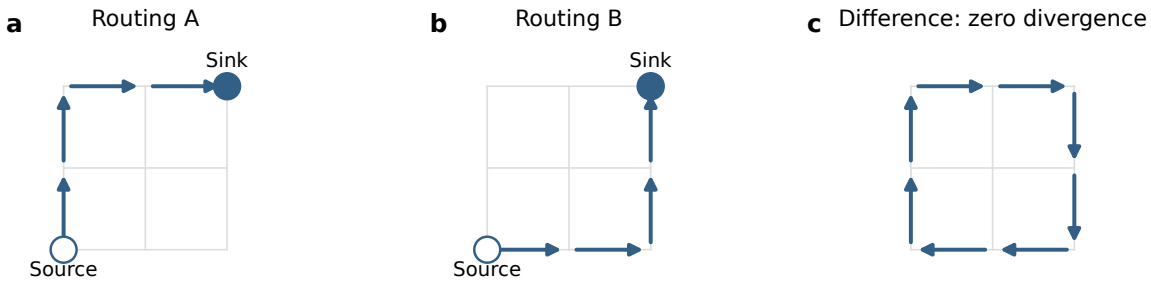


Figure 1: Routing freedom under conservation. a,b, Two routes share a source and sink. **c**, Their difference has zero divergence. Balance fixes the source–sink pattern but leaves routing undetermined.

We approach this question through an effective constitutive relation: a response connecting the source–sink organization of observed flows to the routing component left undetermined by balance. Roads provide a spatial medium that can be measured independently of the mobility values used to estimate the response [19, 20, 21, 22, 23]. The operator asks whether a compact linear mapping can reproduce part of the routing organization, and whether that mapping is specific to the road-defined support rather than to any conservation-compatible subspace of the same size.

Here effective constitutive coupling denotes a reproducible conditional relation between source and routing, together with a testable constraint of an independently measured medium on its structured part. It does not presuppose sufficient closure, unique parameters or a causal response. Beyond whether roads furnish a useful representation, we ask whether their independently measured spatial arrangement constrains the coupling itself.

We estimate this response from trajectories in Porto, San Francisco and Chengdu. UrbanFlow constructs a road-supported routing space, fits a linear source–routing operator on training dates and evaluates it on held-out dates. The analysis links three aspects of the same response: its held-out predictive content, its frequency-resolved structure and its local directional organization.

1 Results

1.1 Effective response

We represent hourly movement as signed crossings of the faces between spatial cells. For a flux field F_t, its discrete divergence $x_t = DF_t$ records the cellwise source–sink balance. Here x_t denotes the divergence of the observed coarse-grained flux, not an independently measured origin–destination demand field. The field separates into a minimum-norm component fixed by that balance and a divergence-free routing component,

$$F_t = D^{\dagger} x_t + F_t^{\circ}, \qquad DF_t^{\circ} = 0. \tag{1}$$

Figure 1 illustrates the physical freedom: different directional configurations can carry the same source and sink. The two components are spatially orthogonal, but their organization can covary across observations. Predicting the routing component from the source therefore tests a statistical relationship between the irrotational and divergence-free sectors of the measured flow.

Road geometry defines an orthonormal basis U_R within the divergence-free space. The road-coordinate state is $y_t = U_R^{\top} F_t^{\circ}$; a fitted response K_R relates this state to the source. The corre-

sponding operator in physical space is

$$\widehat{F}_t^{\circ} = C_R x_t, \qquad C_R = U_R K_R P_x, \tag{2}$$

where P_x restricts inputs to conservation-compatible source patterns. This is a road-supported empirical representation: U_R fixes the routing support and K_R is learned from mobility observations. The medium-constrained operator $C_\theta(W_R)$ introduced in Results 1.4 is a different estimator, not a reparameterization of C_R. Their relation is empirical: the structured road response must add held-out information beyond the freely learned representation (Fig. 5b).

The road space captures most of the observed routing energy in Porto and San Francisco and a substantial share in Chengdu (Supplementary Table S3). This makes the response interpretable in a spatial medium defined independently of the mobility values used to estimate it.

1.2 Road-supported coupling

Operators are estimated on training dates and evaluated on held-out dates. P scores the raw routing field, whereas D tests day-specific departures after removing the training-estimated recurrent hourly pattern (Methods). Figure 2a–c shows that the road-supported operator recovers a meaningful share of both components across the three cities. Porto exhibits the strongest departure response; San Francisco retains a clear response; Chengdu is dominated by recurrent daily structure.

Porto's departure score reaches 0.468 (95% interval 0.355–0.528), and San Francisco's reaches 0.240 (0.188–0.288). A source–routing pairing null, which permutes complete date blocks and refits the operator, produces no draw approaching either observed score. The full-field gain from the same departures is largest in Porto and much smaller in San Francisco and Chengdu. Exact scores, intervals and Chengdu's descriptive small-sample results are given in Supplementary Note S6. Representative physical reconstructions are retained in Supplementary Fig. S6.

The spatial modes expose a complementary difference (Fig. 2d–f). Chengdu's routing is concentrated in a small number of reproducible modes, whereas Porto's organization is distributed across more modes. A compact spatial representation and a source-based response therefore describe distinct aspects of the field.

Full-face ridge and training-only POD representations outperform the road restriction on the same target (Supplementary Table S3). Roads thus provide an interpretable spatial medium rather than an unrestricted approximation of the field.

1.3 Frequency response

The descriptive joint spectra are reported in Supplementary Fig. S1. The frozen operator is evaluated frequency by frequency without fitting a new temporal response (Fig. 3). Because prediction and observation are complex Fourier coefficients, we report a phase-sensitive score,

$$R_m^2 = 1 - \frac{\sum_d \|a_{d,m} - b_{d,m}\|^2}{\sum_d \|a_{d,m}\|^2}, \tag{3}$$

and the pooled signed alignment

$$\widetilde{\gamma}_m = \frac{\sum_d \mathrm{Re}(a_{d,m}^* b_{d,m})}{\sqrt{\sum_d \|a_{d,m}\|^2}\sqrt{\sum_d \|b_{d,m}\|^2}}, \tag{4}$$

where $a_{d,m}$ and $b_{d,m}$ are observed and predicted coefficients on date d (Methods). The response is strongest for the time-independent and low-frequency components, then weakens toward higher

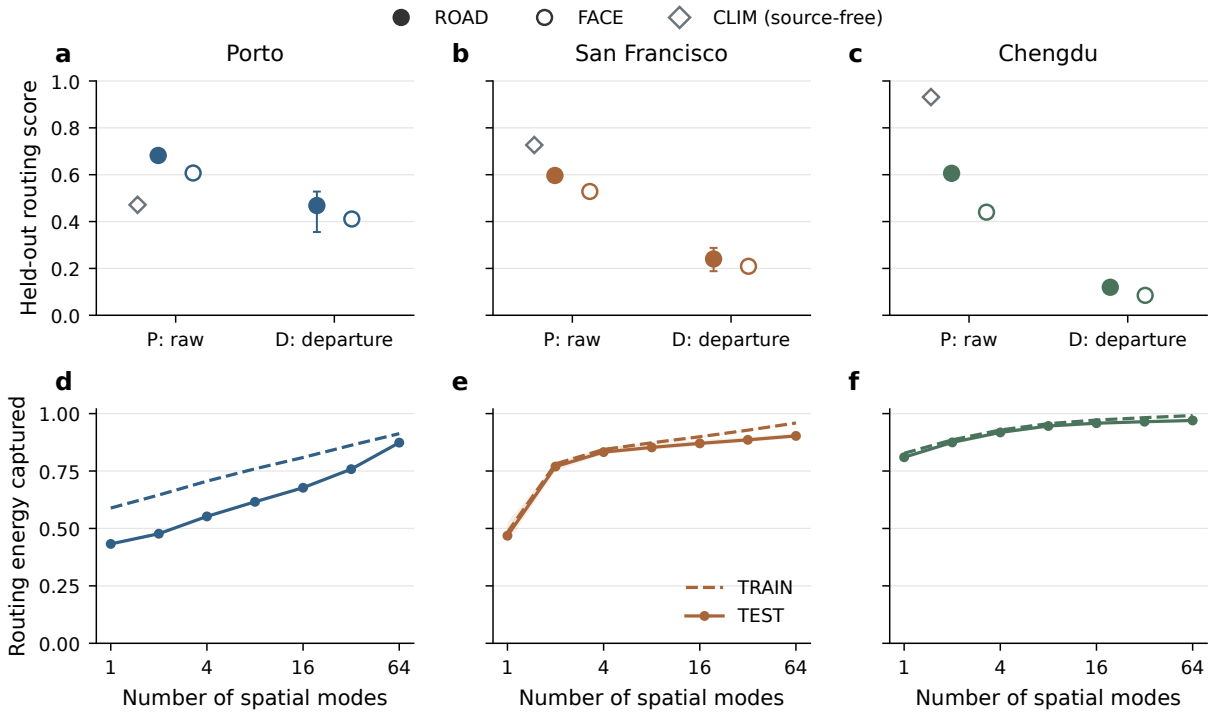


Figure 2: Held-out coupling and spatial organization. a–c, Raw (P) and departure (D) scores on ROAD (filled) and FACE (open) support, pooling TEST errors; diamonds show the source-free P/ROAD climatology. D/ROAD points carry 95% paired-date bootstrap intervals. **d–f**, Energy captured by TRAIN-derived POD modes on TRAIN (dashed) and TEST (solid). San Francisco curves show fold means and ranges. Chengdu uses the 14-date, 06–23 h window. POD capture measures representation, not source-based prediction.

harmonics. The phase-sensitive score falls below zero in several bins in San Francisco and Chengdu. The phase-insensitive magnitude $|\gamma_m|$, shown for comparison, can therefore remain high even when the complex response is not reconstructed accurately.

1.4 Anisotropy and road-medium tests

The fitted response is strongly anisotropic in every city (Fig. 4a–c; full distributions in Supplementary Fig. S4). Its local directional selectivity exceeds that of the road-support tensor and varies substantially across space, and San Francisco's anisotropy map is stable across date folds. Source-covariance weighting preserves this selectivity, so the realized excitation statistics also produce pronounced directional organization. This anisotropy is a property of the fitted response, not a road fingerprint: under source-energy weighting its mean is 0.26, below the 0.32–0.33 of four matched non-road media, and it correlates negatively with road coverage (Spearman −0.18; 999-draw cell permutation $p = 1.00$). Axis agreement with the road support does not exceed a spectrum-matched rotation null (Supplementary Note S8).

The next tests concern a different object: whether the spatial arrangement of the road field enters the response, rather than whether the response tensors align with it. Road geometry identifies a distinctive routing space. Against 999 rank-matched random subspaces of $\ker D$, every random basis leaves more routing energy uncaptured than the road-induced basis (Supplementary Table S4).

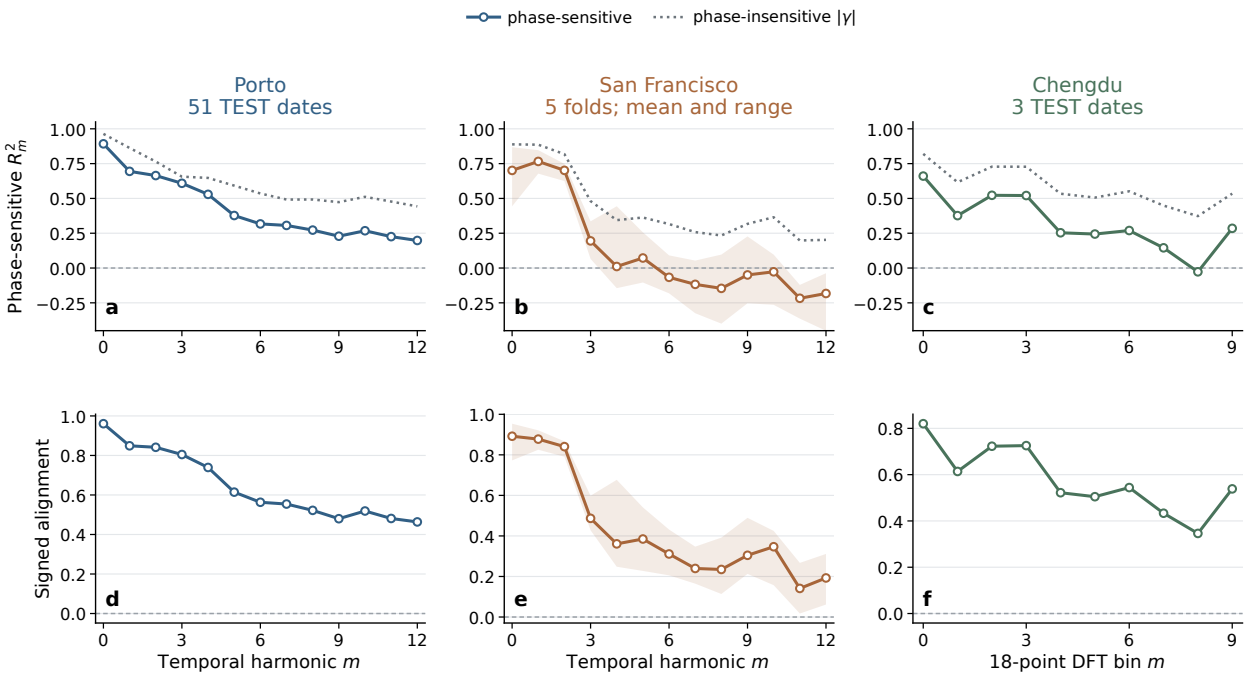


Figure 3: Frequency-resolved response. a–c, Per-bin phase-sensitive R^2_m using per-date complex coefficients. **d–f**, Signed alignment $\widetilde{\gamma}_m$, which pools date-wise complex inner products. The dotted curve in **a–c** is the phase-insensitive magnitude $|\gamma_m|$ from TEST-date mean coefficients. The frozen P-selected operator is used throughout. San Francisco shows fold means and ranges. Porto and San Francisco use 24-point DFTs; Chengdu uses all ten bins of its 18-point observed-block DFT.

The road space is therefore specific within the full aligned observation pool, not merely a convenient choice of coordinates.

At matched nominal capacity, a mobility-inferred medium improves on the road-only medium in every city (Supplementary Table S4; Supplementary Fig. S5). The paired mobility-minus-road differences are positive with narrow date-resampled intervals, while spatial permutation removes the mobility medium's advantage. Road geometry supplies a specific and interpretable support; the learned response additionally contains organization encoded in the mobility data.

The preceding tests concern representation and capacity. To ask whether the road field enters the response as a spatial medium, we fitted a graph operator whose coefficients are learned but whose medium is frozen (Methods). On the D/FACE target (the D frame scored on full FACE support), the road operator reaches 0.062, 0.065 and 0.004 in Porto, San Francisco and Chengdu, against 0.504, 0.258 and 0.126 for an unrestricted FACE-departure ridge (Fig. 5a). A uniform medium produces zero response, and permuting or rotating the road weights collapses the held-out score to at most 0.005. The absolute share is small, but the response is a constrained property of the arrangement rather than a restatement of road coverage.

A directionally matched non-road medium, preserving the exact weight multiset and 26 spatial statistics within the frozen tolerances, never reaches the road score: 0 of 499 draws in Porto and 0 of 199 in San Francisco and Chengdu. The best draws reach 23%, 42% and 47% of the road score. In Porto and Chengdu, a second family that also preserves coarse spatial organization recovers more of the response, at 3.6–4.1 times the directionally matched mean, but the road remains 2.4 and 2.1 times the best coarse draw. San Francisco admits no coarse-matched medium under the frozen

criteria (0 of 199), so its evidence supports directional specificity but not a fine-scale attribution. These counts are nominal: participation-ratio effective sizes are 148/93/175 for Null-1 and 5.6/–/12.4 for Null-2, so the coarse-family tail bounds the constructed family rather than supplying a calibrated p-value.

The road medium also adds information beyond a flexible learner. Used as a prior under an equal-capacity residual ridge, it raises held-out R^2 by 0.0116, 0.0207 and 0.0023 in the three cities (Fig. 5b), corresponding to relative residual-SSE reductions of 2.27%, 2.65% and 0.26%. The 20 directionally matched priors change held-out R^2 by between -0.0013 and $+0.0003$. The road increment is positive on every Porto TEST date and in three of four blocked Porto folds, and it is largest when TRAIN is reduced, consistent with a structural prior that substitutes for data. It is nevertheless modest, and the learned correction has no reproducible response direction (median half-TRAIN cosine 0.20–0.46). We therefore treat it as an informative constraint, not an identified operator.

Independently measured static urban structure does not substitute for the road medium. We repeated the same frozen operator family, split and score on Porto with population density, built-up surface and built land-use fractions (transport classes excluded) from external GIS layers, alone and in additive or multiplicative combination with the road field. Each field on its own explains at most 0.003 of held-out FACE variance, whereas the road medium reaches 0.0619. Adding the road medium to any field raises held-out R^2 by 0.0200–0.0384; adding any field to the road medium lowers it by 0.0239–0.0411. The complementarity is therefore one-directional, and the best additive joint medium (0.0380) remains below the road medium alone. The multiplicative land-use composition does not clear the pre-registered 0.01 margin (+0.0023), so the conclusion is stated for the additive composition and the measured fields only; all media remain weak in absolute terms (Supplementary Table S6).

1.5 Model comparison

Table 1 reports the modern conditional-reconstruction comparison on the D/FACE target under one pooled ratio-of-sums score. The linear ridge and POD reference remains the strongest source-only model in all three cities. The original road-supported UrbanFlow operator is also included; it reaches 0.411, 0.209 and 0.085 on the same D/FACE target. In Porto, ridge reaches 0.5044 and the best frozen neural arm, an MLP with road channels, reaches 0.4473; San Francisco and Chengdu reproduce the ordering (best neural 0.2319 versus 0.2582, and 0.0125 versus 0.1258). Road channels help some architectures and hurt others, but no neural arm closes the gap. Re-selecting the MLP, U-Net and Fourier configurations on VAL alone leaves the ordering unchanged: the best arm, an MLP, reaches 0.4404 (s.d. 0.0208) against 0.5044 for the linear ridge, while the U-Net and Fourier arms reach 0.1589 and 0.1414. The gap is therefore not an artefact of the frozen neural configurations. Within the same linear family, adding the frozen E2 road-medium response as a feature block raises the ridge to 0.5405 (+0.0361, 95% interval 0.0283–0.0418, positive on all 51 TEST dates); adding two matched null blocks to that road block raises VAL but lowers TEST to 0.5343. We read this increment as a structural inductive bias and a regularization effect within the tested linear class, not as an expanded linear function class or new causal information. GraphGPS was run only in Porto; the San Francisco and Chengdu replication uses the remaining architectures. The comparison uses five seeds per neural arm on Porto and is specific to the tested capacities, sample size and 300-epoch protocol. Table 1 is D/FACE only: the original P/FACE model, Graph WaveNet and the gravity/loading diagnostics are retained in Supplementary Table S2 and are not ranked with these values.

Table 1: Model comparison.

Model (D/FACE)	Porto	San Francisco	Chengdu
Linear reference and road-structure contrast			
Linear ridge	0.5044	0.258	0.126
POD regression	0.504	0.267	0.126
UrbanFlow (road-supported)	0.411	0.209	0.085
Ridge + E2 road block	0.5405	–	–
Ridge + E2 and two matched blocks	0.5343	–	–
Neural arms, frozen configurations			
MLP, source	0.421 (0.010)	0.070 (0.004)	−0.070 (0.005)
MLP, source + road	0.447 (0.019)	0.098 (0.006)	−0.019 (0.013)
U-Net, source	0.183 (0.015)	0.198 (0.023)	0.012 (0.004)
U-Net, source + road	0.162 (0.005)	0.232 (0.012)	0.012 (0.003)
GraphGPS, source	0.158 (0.015)	–	–
GraphGPS, source + road	0.019 (0.024)	–	–
Fourier operator, source	0.029 (0.010)	0.142 (0.006)	0.0003 (0.0002)
Fourier operator, source + road	0.110 (0.011)	0.207 (0.006)	0.007 (0.002)
Neural arms, VAL-reselected (Porto)			
MLP, source + road	0.4404 (0.0208)	–	–
U-Net, source + road	0.1589 (0.0052)	–	–
Fourier operator, source + road	0.1414 (0.0153)	–	–

Pooled TEST R^2 on the full divergence-free departure. Neural entries are means (s.d.) over five seeds; linear rows are a single frozen fit. The VAL-reselected rows use the configuration selected on VAL alone and are reported at the precision of their frozen outputs; they share the target, splits and 300-epoch budget but are not the frozen Task H configurations. The road-block rows add the frozen E2 response, and the matched-block row adds two same-size null response blocks. Dashes mark configurations not run. The zero-departure reference has $R^2 = 0$ by construction; negative entries are worse than predicting the TRAIN phase mean. The UrbanFlow row evaluates the original road-supported operator on the same frozen D/FACE target; it is not a re-fit. The P/FACE model, Graph WaveNet, gravity and loading diagnostics are in Supplementary Table S2; the two frames are not mixed. Full training configurations and paired-date contrasts are in Supplementary Note S10.

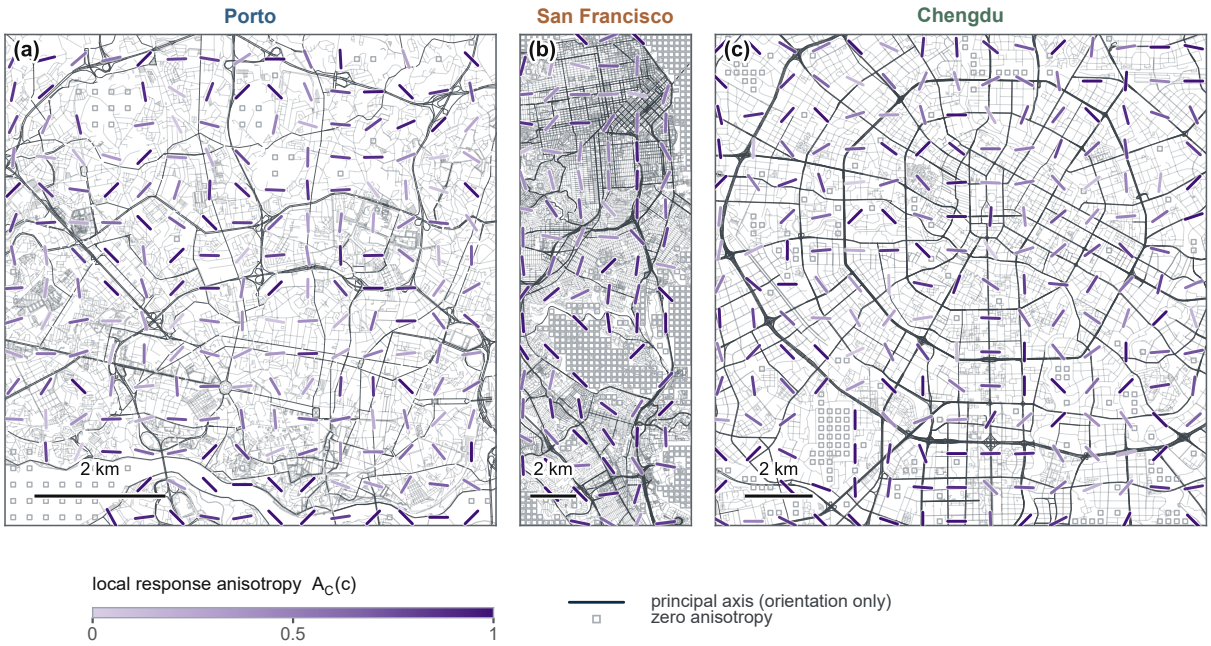


Figure 4: Anisotropic empirical response. a–c, Local response anisotropy A_C on 250 m regular interior lattices at true aspect (San Francisco: five-fold consensus). Colour shares 0–1; segments show axial directions for $A_C > 0.02$ and hollow squares mark zero anisotropy. Streets are © OpenStreetMap contributors. The anisotropy is a property of the fitted response, not a road fingerprint: axis agreement with the road support does not exceed a spectrum-matched rotation null (Supplementary Note S8).

2 Discussion

2.1 A constitutive description of urban movement

The identified operator adds an empirical relation to the Hodge decomposition: source–sink organization and divergence-free routing are orthogonal at one instant, yet covary across observations. The evidence supports a partial, observation-conditioned effective coupling with road-specific structural constraints, not a sufficient law. The response is strongest in Porto, weaker in San Francisco and small in Chengdu, where recurrent structure already accounts for much of the field. The comparison with unrestricted representations further shows that the road-supported object is interpretable rather than merely accurate.

The lifted operator C_R maps a distributed source pattern to a directional routing response. Its columns describe model-implied responses to compatible source directions, while its local tensors expose spatially varying directional selectivity. The phase-sensitive frequency analysis and the anisotropy analysis therefore characterize the same response in time and space. Together, they turn a fitted matrix into an interpretable description of urban movement at the observation scale.

2.2 Urban anisotropy and the role of roads

Anisotropy separates the geometry available to movement from the directional response expressed through it. The fitted response is locally selective rather than organized around one city-wide axis. Neither the orientation nor the magnitude of these tensors is diagnostic of road alignment: axis agreement with road support does not exceed a spectrum-matched rotation null, and the source-

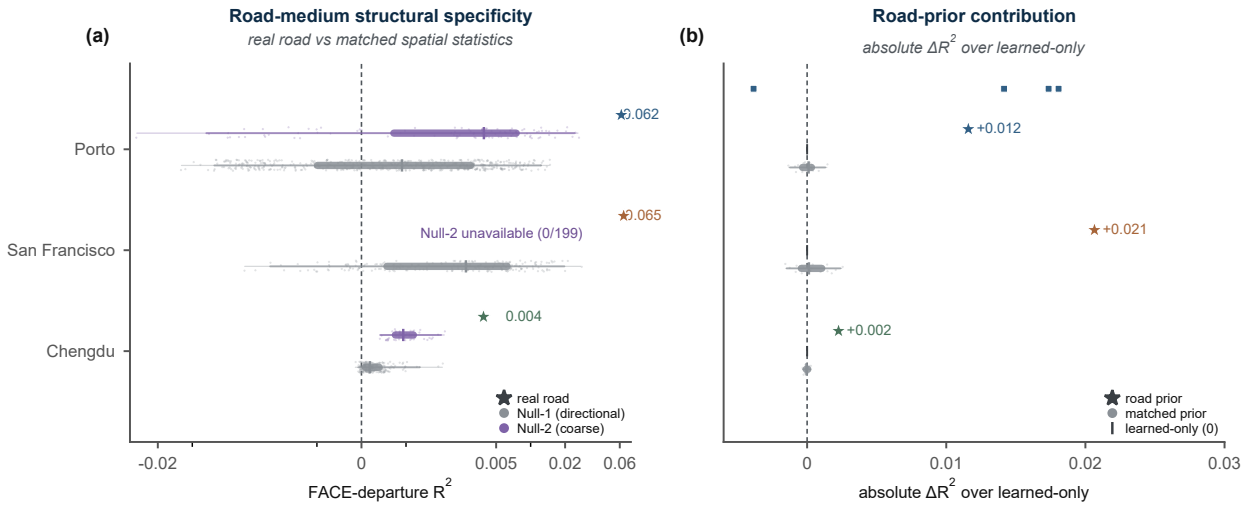


Figure 5: Road-medium structural specificity and prior contribution. **a**, FACE-departure R^2 for the real road operator (stars), directionally matched Null-1 (gray) and coarse-matched Null-2 (purple) distributions. Points, boxes and whiskers show draws and percentiles; SF has no constructible Null-2 (0/199). The a panel uses a symmetric-log axis (linear threshold 0.002) because the null distributions include small negative values. Nominal/effective draw counts and the Monte Carlo floor are in Supplementary Note S9. **b**, Absolute ΔR^2 of the road prior and 20 matched priors over the learned-only reference; the four squares above the Porto row are the four blocked-fold increments, with one negative block (3/4 positive). Rank-matched and matched-capacity controls are retained in Supplementary Table S4.

weighted response anisotropy (0.26) lies below that of four matched non-road media (0.32–0.33). We therefore interpret anisotropy as a property of the fitted operator on its identified source span, not as independent evidence of road-aligned response. This connects street configuration [24, 25, 26] to a measured operator.

Roads play two roles in this framework. As a geometric support, they identify a specific divergence-free space; as a response medium, they do not exhaust the organization encoded in the mobility data. The matched-capacity comparison makes this distinction explicit. A road-only medium is not sufficient, but the road-supported operator remains interpretable and specific. The appropriate conclusion is a road-supported effective coupling, not a road-determined constitutive law.

The matched-medium tests sharpen the word *effective*. A non-road medium that matches the road's weight distribution, directional correlations and, where constructible, coarse spatial organization does not reproduce its held-out response. The road prior also adds a small but reproducible increment over a matched learner, while 20 matched priors do not. These results support a road-specific partial constraint, not a sufficient closure or a mechanism. The claim is deliberately limited to the tested surrogate families and the frozen observation regimes. Independently measured population, built-up area and built land-use fields do not substitute for the road medium: each explains at most 0.003 of held-out variance on its own, adding the road medium to any of them raises the score by 0.020–0.038, and adding any of them to the road medium lowers it by 0.024–0.041. Within the same linear family, a frozen road response block adds a transferable increment of 0.036, whereas two matched null blocks do not. The road structure therefore carries practical information value in the tested models; this is not a mechanism claim.

A structured 14-atom operator fitted in Porto provides the complementary negative result (Supplementary Note S11). It improves the D/FACE score by 0.022, but matched surrogate media reproduce the gain, the design is numerically singular, and the selected coefficients are not identified physical road parameters. Post-hoc attribution likewise fails to distinguish real from permuted road fields. This experiment tests one estimator, one dictionary and one city; it does not weaken the road-specific evidence above, and we do not present it as a new explanation method.

Taken together, the evidence supports a partial, observation-conditioned effective coupling with road-specific structural constraints. Figure 2 supplies the empirical coupling; the matched-medium and independent-field tests supply the road-specific constraint; the anisotropy tests deny that road geometry fixes its direction or magnitude; the regime splits deny demonstrated invariance; and the structured dictionary denies unique physical identification. The object is predictive and structural, not a sufficient or regime-invariant constitutive law.

2.3 Scope and further tests

Several boundaries define the next experiments. Signed source balance compresses movement: different OD pairings and opposing flows can produce the same net balance. Chengdu has only three held-out dates in the primary protocol, so its departure interval and pairing null are descriptive; its matched-capacity interval uses four alternate-hour dates. The rank-matched null does not compare a rebuilt or rewired road graph. The E/F/G tests permute road weights on the fixed graph complex; they are medium-weight counterfactuals, not topology interventions. A decisive structural intervention would compare a rebuilt network at the same rank, input and target. Nonlinear responses, additional state variables and longer memory are further testable extensions. The road rasters carry no acquisition date or OpenStreetMap snapshot identifier: they postdate the mobility windows by roughly 12–13 years in Porto and 18 years in San Francisco, while the Chengdu date is unrecorded. The results therefore describe a road field measured later than the movement, not the network of the mobility era.

Excitation-regime stability is a further boundary. Splitting TRAIN dates by source activity or by spatial type and refitting within each condition did not improve held-out scores over the shared operator (paired VAL differences -0.009 to -0.064), and the cross-condition response similarity (cosine 0.64–0.66) was indistinguishable from capacity-matched random date subsets (0.60–0.66). A second operator family preserved the response shape across conditions (cosine ≥ 0.998) but lost 10–25% of its transfer R^2. We therefore do not claim that the coupling is invariant across source regimes; these splits condition the data and are not interventions.

The operator is an effective description of the sampled city, not an intervention model. Sources and targets come from the same trajectories, and each city's operator is fitted separately. Forecasting from independently available demand, transfer between cities and model-implied effects of infrastructure changes require further validation. The identifiable content is the response on excited source directions and its reproducible out-of-sample contribution. Unique physical medium parameters, a single constitutive matrix and regime-invariant material properties are not identified.

3 Methods

3.1 Data and discrete fields

Porto uses ECML/PKDD taxi trajectories [27]: 252 dates, a chronological TRAIN/VAL/TEST split of 151/50/51 dates and a 32×32 lattice. San Francisco uses Cabspotting [28]: 15 dates, five

blocked folds of 9/3/3 dates and a 32×88 lattice. Chengdu uses DiDi ride-hailing trajectories [29] with 30-second sampled fixes and a 60×60 lattice. Its P/D, coherent-response, anisotropy and spectral analyses use a 14-date 8/3/3 subset from the continuous-observation regime. The matched-capacity medium sweep uses a separate chronological 10/4/4 split whose four held-out dates are 26–29 August 2014; these dates belong to the alternate-hour delivery phase, with nine alternating recorded hours per date. The structural null is evaluated on the 19 valid dates of the aligned grid. These protocols are related but not interchangeable. The hourly phase sets are 00–23 h for Porto/San Francisco and 06–23 h for Chengdu. Each city is estimated separately. Observation regimes and frozen date lists are specified in Supplementary Note S1 and Supplementary Table S1.

Successive trajectory segments are walked through the aligned nominal 250 m cells. Signed crossings of each interior face are accumulated in 60-minute windows to form $F_t \in \mathbb{R}^{n_f}$. No map matching is applied. For the cell–face incidence matrix $D \in \mathbb{R}^{n_c \times n_f}$,

$$x_t = DF_t, \quad P_0 = I - D^\dagger D, \quad F_t^\circ = P_0 F_t, \quad P_x = DD^\dagger. \tag{5}$$

The routing target is calculated from the same crossing field as the source. The Laplacian solve is gauge-pinned. The source is an observed coarse-grained divergence, not an independently measured OD demand field.

The archived footprint mask is the union of cells with non-zero density over all dates, (N > 0).any(0). It is transductive and is used only for analysis and display; it is not an input to D, U_R or K_R, which depend on the grid, study domain and road registration.

3.2 Road space and response estimation

The drivable OpenStreetMap layer [30] is registered to the study complex. Let B_R be its node–edge incidence, A_R the edge-to-face lift and Π the registration map satisfying $DA_R = \Pi B_R$. A cycle basis Z_R satisfies $B_R Z_R = 0$. The Gram eigendecomposition of $A_R Z_R$ gives

$$U_R = \mathrm{orth}(A_R Z_R), \qquad U_R^\top U_R = I, \qquad DU_R = 0, \qquad y_t = U_R^\top F_t^\circ. \tag{6}$$

Geometry, study domain and road registration determine U_R. Supplementary Note S2 gives the registration conventions.

For any source or target v_t, TRAIN phase means and departures are

$$\bar{v}_\phi = \frac{1}{n_{\mathrm{tr},\phi}} \sum_{t \in \mathrm{TRAIN}:\phi(t)=\phi} v_t, \qquad \delta v_t = v_t - \bar{v}_{\phi(t)}. \tag{7}$$

Stack $\delta x_t^\top$ and $\delta y_t^\top$ as the rows of X and Z. Ridge estimation yields

$$\begin{aligned} K_R^\star &= \arg\min_K \{\|Z - XK^\top\|_F^2 + \lambda \|K\|_F^2\} \\ &= Z^\top X (X^\top X + \lambda I)^{-1}, \qquad C_R = U_R K_R^\star P_x. \end{aligned} \tag{8}$$

We use $\lambda = \kappa s_{\max}(XX^\top)$ and select κ on VAL from the grid specified in Supplementary Note S2. The archived values are $\kappa = 0.01$ in Porto, 0.1 in San Francisco and 0.1 in Chengdu's D frame; the Chengdu P-frame sensitivity uses 0.001. Comparison arms select regularization in the specified scoring frame. Full-face ridge fits F_t° and projects predictions with P_0; POD estimates its output basis from TRAIN departures and selects rank and regularization using FACE validation error.

3.3 Scoring

Define $T_t = U_R^\top F_t^\circ$ on ROAD support and $T_t = F_t^\circ$ on FACE support. Let B_{P} and B_{D} denote the fitted maps on that support. Both use TRAIN departures; P selects on raw VAL inputs/targets, while D selects on VAL departures using TRAIN means. Predictions have no fitted intercept:

$$\begin{aligned} R^2_{\mathrm{P}} &= 1 - \frac{\sum_{t\in\mathrm{TEST}} \|T_t - B_{\mathrm{P}} x_t\|^2}{\sum_{t\in\mathrm{TEST}} \|T_t\|^2}, \\ R^2_{\mathrm{D}} &= 1 - \frac{\sum_{t\in\mathrm{TEST}} \|\delta T_t - B_{\mathrm{D}} \delta x_t\|^2}{\sum_{t\in\mathrm{TEST}} \|\delta T_t\|^2}. \end{aligned} \tag{9}$$

Numerators and denominators are pooled across TEST observations and folds within each city. The denominators are target energies. The source-free reference is $\bar{T}_{\phi(t)}$, reported in Supplementary Table S3.

3.4 Structural tests

For any orthonormal routing basis U, the uncaptured-energy statistic is

$$E(U) = 1 - \frac{\sum_t \|U^\top F_t^\circ\|^2}{\sum_t \|F_t^\circ\|^2}, \qquad \Delta E_b = E(U_b) - E(U_R). \tag{10}$$

Let K_D be an orthonormal basis of $\ker D$. Each null draw uses $G_{ij} \overset{\mathrm{iid}}{\sim} N(0,1)$, $Q = \mathrm{orth}(G)$ and $U_b = K_D Q$ at rank $\mathrm{rank}(U_R)$. The test uses the full aligned observation pool and no source-response fit; this pool differs from the frozen P/D comparison pool.

The second test compares road, mobility-inferred and spatially permuted mobility media at equal nominal block capacity R. Each per-face medium is reduced to the same R on nested spatial partitions before mean-1/floor normalization; $R = 1$ gives the constant-medium reference. Permutation preserves values, partition and nominal block count. The archived statistic is the equal-date mean of

$$R^2_{\mathrm{cond},d} = 1 - \frac{\sum_{t\in d} \|T_t - \widehat{T}_t\|^2}{\sum_{t\in d} \|T_t - \langle T\rangle_d\|^2}. \tag{11}$$

Supplementary Table S4 reports the paired-date interval for mobility minus road at $R = n_{\mathrm{int}}$, the number of interior faces (1984, 5512 and 7080); Supplementary Fig. S5 reports each medium's gain over the constant medium. Chengdu uses the archived all-motion channel and a four-date held-out protocol. This date-centred statistic is separate from Eq. 9. Source files and partitions are specified in Supplementary Note S4.

3.5 Road-medium operator

To test whether the spatial arrangement of the road field enters the response, we use a graph operator whose coefficients are fitted but whose medium is frozen. Let B_2 be the edge-to-plaquette boundary operator and

$$L_1 = D^\top D + B_2 B_2^\top, \qquad \tilde{L}_1 = \frac{2L_1}{\lambda_{\max}(L_1)} - I. \tag{12}$$

With $W_R = \mathrm{diag}(w_e)$ the road weight on interior faces, normalized to unit mean on its positive support with floor 10^{-3}, the operator family is

$$C_\theta(W_R) = P_0 \left[\sum_{k=0}^{K} a_k \frac{W_R T_k(\tilde{L}_1) + T_k(\tilde{L}_1) W_R}{2} \right] D^\dagger, \tag{13}$$

where T_k are Chebyshev polynomials. The response is linear in $\theta = (a_0, \ldots, a_K)$; K is selected on VAL from $\{0, 1, 2, 3, 4, 6\}$ and the ridge penalty follows the same relative grid as Eq. 8. The construction satisfies $DC_\theta = 0$ for every medium, and a uniform medium gives $C_\theta(cI) = 0$ exactly. Nonzero response therefore requires spatial heterogeneity. The selected order reaches the pre-registered upper limit $K = 6$ in all three cities. This shows that the VAL search did not settle below the boundary; it does not establish that a wider family would improve TEST performance. The grid was not widened after TEST.

The structural comparison uses the FACE-departure target,

$$R^2_{\mathrm{D,FACE}} = 1 - \frac{\sum_{t \in \mathrm{TEST}} \|\delta F^\circ_t - \delta \widehat{F}^\circ_t\|^2}{\sum_{t \in \mathrm{TEST}} \|\delta F^\circ_t\|^2}, \tag{14}$$

with δ defined by the TRAIN phase means in Eq. 7. This is the D/FACE target. It differs from the P/FACE score in Table 1 in the target (departure versus raw) and from the ROAD-support D score in Fig. 2 in the scoring space (full FACE versus U_R); it is not a separate scoring convention.

Matched media. Two surrogate families preserve properties of W_R while destroying its arrangement. The directionally matched family preserves the exact weight multiset and 26 lag, smoothness and cross-family statistics within pre-registered tolerances; each accepted draw also has small correlation with the road field and no more than 25% top-decile overlap. The coarse-matched family additionally preserves the road's $\sigma = 2$ cell-scale projection by permuting within 20 equal-count strata, and accepts a draw only when the fine residual is weakly correlated with the road field. Every candidate is fitted with the same operator family, split and selection rule. Tail estimates use the Monte-Carlo floor $1/(N+1)$ and are read as comparisons with the constructed family, not as permutation p-values.

Prior control. The road operator also supplies a prior $\widehat{F}^\circ_{\mathrm{prior}} = C_\theta(W_R)\delta x$; an equal-capacity reduced-rank ridge learner is fitted to the residual $\delta F^\circ - \widehat{F}^\circ_{\mathrm{prior}}$ with a frozen rank and penalty grid selected on VAL. The reported absolute increment is $\Delta R^2 = R^2_{\mathrm{joint}} - R^2_{\mathrm{learned}}$, and the relative gain is

$$g = \frac{\mathrm{SSE}_{\mathrm{learned}} - \mathrm{SSE}_{\mathrm{joint}}}{\mathrm{SSE}_{\mathrm{learned}}}. \tag{15}$$

These quantities are not interchangeable. The control repeats the procedure with 20 directionally matched priors. Porto is additionally evaluated in four contiguous date blocks separated by a five-date embargo, with inner 75/25 TRAIN/VAL selection; this is blocked internal validation, not an external test. Full matching tolerances, effective sample sizes and block results are in Supplementary Note S4.

3.6 Anisotropy

For each regular interior cell, L_c sums signed face flux into two physical components. Define

$$\begin{aligned}\Sigma_C(c) &= L_c C_R C_R^\top L_c^\top,\\ \Sigma_U(c) &= L_c U_R U_R^\top L_c^\top,\\ A(c) &= \begin{cases}(\lambda_1(c)-\lambda_2(c))/(\lambda_1(c)+\lambda_2(c)), & \lambda_1(c)+\lambda_2(c)>0,\\ 0, & \lambda_1(c)+\lambda_2(c)=0.\end{cases}\end{aligned} \tag{16}$$

Here $\lambda_1 \geq \lambda_2$ are the eigenvalues of the relevant tensor. The principal eigenvector defines an axial orientation θ modulo π. San Francisco uses the per-cell mean anisotropy and nematic mean orientation over five folds. We also evaluate the source-covariance-weighted response tensor

$$\Sigma_{\text{response}}(c) = L_c C_R \Sigma_{xx} C_R^\top L_c^\top, \tag{17}$$

with Σ_{xx} the TRAIN covariance of source departures. Structural-axis agreement is evaluated on regular interior cells for which both anisotropies are nonzero, using the frozen joint-anisotropy weighting

$$G_\theta = \frac{\sum_c A_C(c) A_U(c) \cos[2(\theta_C(c)-\theta_U(c))]}{\sum_c A_C(c) A_U(c)}. \tag{18}$$

To test whether this agreement exceeds what the shared road subspace and operator spectrum imply, write $K_R = Q\Lambda V^\top$ and replace Q by a Haar-random orthonormal frame Q_0 in the road output space, $K_0 = Q_0 \Lambda V^\top$. We recompute G_θ for 999 draws.

Only regular interior cells enter the anisotropy statistics; their metric $L_c L_c^\top = 2I$ is isotropic. Metric and rotation checks are specified in Supplementary Note S4.

3.7 Spectra and coherent response

For the joint spectrum in Supplementary Fig. S1, each TRAIN date is transformed separately, retaining its mean. For field $v_{d,j}$ and observed block length N,

$$\begin{aligned}Z_{d,m} &= N^{-1/2} \sum_{j=0}^{N-1} v_{d,j} e^{-2\pi i m j/N}, \qquad f_m = m/N,\\ P_m &= w_m \langle \|Z_{d,m}\|^2 \rangle_d, \qquad J_{m,k} = \frac{P_m}{\sum_l P_l} \frac{s_{m,k}^2}{\sum_a s_{m,a}^2}.\end{aligned} \tag{19}$$

$s_{m,k}$ are singular values of the date-by-space coefficient matrix. The weights are $w_0 = w_{N/2} = 1$ and $w_m = 2$ otherwise. Porto/San Francisco use $N = 24$; Chengdu uses $N = 18$ with no night-time imputation. Frequencies are cycles per observed hour. San Francisco averages P_m and within-frequency mode shares separately across folds before multiplication. The first eight modes are displayed after full-spectrum normalization. Spatial POD curves in Fig. 2 use a raw TRAIN routing basis.

For the frequency-resolved response, each TEST date is transformed with the same unitary DFT used for the descriptive spectrum: 24 points in Porto/San Francisco and 18 points in Chengdu. Let $a_{d,m}$ and $b_{d,m}$ be the observed and predicted road-coordinate coefficients on date d at bin m. We report

$$R_m^2 = 1 - \frac{\sum_d \|a_{d,m} - b_{d,m}\|^2}{\sum_d \|a_{d,m}\|^2}, \qquad \widetilde{\gamma}_m = \frac{\sum_d \mathrm{Re}(a_{d,m}^* b_{d,m})}{\sqrt{\sum_d \|a_{d,m}\|^2}\sqrt{\sum_d \|b_{d,m}\|^2}}, \tag{20}$$

where the signed alignment pools date-wise complex inner products and energies across TEST dates. The phase-insensitive magnitude $|\gamma_m| = |a_m^* b_m|/(\|a_m\|\|b_m\|)$ uses TEST-date mean coefficients and is reported for comparison. The P-selected operator is fixed. All ten single-sided bins are evaluated in Chengdu.

3.8 Baselines and uncertainty

The neural comparison contains a residual MLP, a U-Net [**?**], a Fourier neural operator [**?**] and GraphGPS [**?**]. Graph WaveNet uses the frozen splits, a 300-epoch budget and a four-point VAL-selected grid, with phase-centred TRAIN data and raw VAL/TEST data (P/FACE). Supplementary Table S2 uses archived pooled per-date errors; sealed seed robustness and the archived Porto paired contrast are reported in Supplementary Note S5. Graphormer's reduced-budget run is excluded. Configurations and outputs are indexed in Supplementary Note S12.

Gravity estimates OD flows from TRAIN margins and distance, selecting friction on VAL. Its overlap is CPC $= 2\sum_{ij}\min(\widehat{Q}_{ij}, Q_{ij})/(\sum_{ij}\widehat{Q}_{ij} + \sum_{ij} Q_{ij})$ at 4×4-cell aggregation. AON uses time-shortest paths, road costs and a 150 m snap threshold; gravity loading retains the top 150,000 zone pairs. Scaled diagnostics fit α^* on TEST; the fixed-amplitude arm uses $\alpha = 1$. Chengdu OD uses maximal occupied runs under the frozen cleaning rules. Registration and loading conventions are specified in Supplementary Note S5; no radiation or user-equilibrium score is estimated.

Primary D intervals use 10,000 resamples of held-out dates within each fold and the 2.5/97.5 percentiles; faces are never resampled. The source–routing pairing null permutes complete date blocks within TRAIN, VAL and TEST separately and refits the operator in each of 199 draws. Neural contrasts use 9,999 date resamples with Holm adjustment within city/support; matched-capacity contrasts use 10,000 paired-date resamples. Structural nulls use 999 draws. Chengdu's pairing null is descriptive because it has only three held-out dates; its equal-capacity interval resamples four dates.

Data availability

Porto and San Francisco use the cited public datasets; Chengdu uses the archived DiDi ride-hailing sample [29]. Data versions and checksums are recorded in the repository data manifest. Road data derive from OpenStreetMap and retain the OpenStreetMap contributors attribution. Frozen splits and observation regimes are tabulated in the supplementary material. The public export contains no individual-level trajectory records; redistribution status and any external-data requirements for derived fields are recorded in its data manifest.

Code availability

The analysis repository separates figure rebuilding from frozen products, core-score recomputation and end-to-end raw-trajectory reconstruction. The matched-capacity modules, phase-sensitive frequency metrics, anisotropy rotation null and figure generators are included. The matched-capacity sweep still requires external aligned fields, date-freeze files and GIS rasters, which are listed with hashes in the data manifest; no raw trajectories are included. Large prediction files and original training artifacts that are not in the export are identified rather than represented as included. Execution requirements and checksums are documented in the repository manifest. Public repository location: https://github.com/Wangwangw233/paper-urbanflow.

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